

youkined from <https://github.com/Nicicalu/OST-CN2-Spik>

OSPF (IGP)
Equal Cost Multi-Path (ECMP) load balancing, fast convergence, widely used.
Uses IP Port 89 on L3. Uses 224.0.0.5 (all routers), 224.0.0.6 (all DR,BDR)
DR Election: Highest IF prio → If tie, highest Router ID. Priority 0 = ineligible for DR/BDR. No preemption. DR = same rules, second best candidate
Virtual links: Cannot go through more than one other area, can only run through standard non-backbone areas. (eg. not over stubby areas)

ROUTER OPERATION
– Establish neighbor adjacencies and exchange LSAs
– Build the Link-State Database (LSDB)
– Run Dijkstra's SPF algorithm:
– **Intra-area change:** SPF recalculation
– **Inter-area change:** No SPF needed → ABR handles updates
– Build the routing table from SPF results

ROUTE TYPES
O: Intra-area: Local area → Local area. Type 1 and 2 LSAs
E1: Intra-area: Local area → Neighboring area. Via LSA 3 through backbone
O E1 or O E2: External: Local area → External (eg. BGP), from ASBR
O E1: Cost = internal + external metrics
O E2: Cost = external metrics. Default external route for OSPF
Cost calculation: Reference Bandwidth(def=100 Mbps)/Interface Bandwidth

AREAS
AS Split into **areas** with 32-bit Area ID (e.g., 0.0.0.0 = Area 0)
Backbone Area (Area 0): The OSPF domain must connect to all other areas (directly/virtual links), must be contiguous (no disjointed segments), should not contain end-user networks, summarizes topology of other areas
Non-Backbone Areas: Connect end users and local resources, All inter-area traffic must transit the backbone

ROUTER TYPES
Area Border Router (ABR): Connects two or more OSPF areas, must have 1 interface in backbone, 1 OSPF DB per Area
Internal Router (IR): All interfaces belong to the same area (non-backbone)
Backbone Router (BR): At least one interface in Area 0, includes ABRs and routers internal to the backbone
AS Boundary Router (ASBR): Connects to external AS/Network (e.g., BGP), Advertises external routes into OSPF, can be in backbone or non-backbone

DESIGN RULES
Rule 1: Backbone (Area 0) must be contiguous → no partitions allowed
Rule 2: Every non-backbone area must connect to Area 0

LINK STATE ADVERTISEMENT (LSA) TYPES
(1) Router-LSA: All routers list directly connected links (so all outgoing inter-faces with state and cost) (**intra-area**)
(2) Network-LSA: **DR** lists all routers and DR in broadcast/multi-access network (**intra-area**)
(3) Summary-LSA: ABR advertises networks from other areas, flooded in all links that are not "totally stubby" (**inter-area**)
(4) ASBR-Summary: **ABR** advertises path to ASBRs. **ASBR** sends Type 1 + external bit, ABR converts to Type 4 and floods (**inter-area**)
(5) AS-External-LSA: **ASBR/ABR** advertises external routes (e.g., BGP); flooded to all non-stub areas → only normal areas (**inter-area**)
(7) NSSA External: Like Type 5 but inside **NSSA**. **Converted to Type 7 by ABR**. Shown in routing table as **[O N1 or O N2]**

AREA TYPES (ALLOWED LSA TYPES)
Stubby = Blocks external LSA [No 5], Totally = Blocks inter-area LSA [No 3/4]
Reference: (e.g., 0.0.0.0) from **ABR [DR ABR]**, Cannot contain **ASBR [No ASBR]**
Standard (1,2,3,4,5): **Total**
Stub Area (1,2,3): **[No 4/5] [DR ABR] [No ASBR]**
Totally Stubby Area (1,2): **[No 3/4/5] [DR ABR] [No ASBR]**
Not-So-Stubby Area (1,2,3,7): **[No 4/5]**, no DR, allows 1 ASBR **[LSA 7 → 5]**
Totally NSSA (1,2,7): **[No 3/4/5] [DR ABR] [LSA 7 → 5]**

PACKET TYPES
(1) Hello: Usually Multicast. Discover, maintain, and verify neighbors. Forms adjacencies. Election of DR/BDR in broadcast networks. Contain network mask (sending routers' IF), Hello interval (p2p,broadcast: def=10s), Options, Priority (election), Router dead interval (def=40s), DR/BDR IF, Neighbors.
(2) Database Description (DBD/DD): Unicast. Exchange summaries of LSAs during adjacency formation (headers only). Contains IF, Max MTU, Options, IAMS/Bits (Initial, More, Master-slave bit), DD Seq, Num., LSA Header
(3) Link State Request (LSR): Unicast. Request specific LSAs. Contains Link State Type (router/net), Link State ID, Advertising Router (sender address)
(4) Link State Update (LSU): Multi/Unicast. **Implicit ack.** Flood new/updated LSAs. Answer to LSR (unicast). Contains Nr. of LSAs, full LSAs information.
(5) Link State Acknowledgment (LSAck): Multi/Unicast. **Explicit ack.** Confirms receipt of LSAs for reliable flooding. Acks may be grouped together.

SUB-PROTOCOLS
Hello Protocol: Neighbor discovery, Parameter negotiation. Maintain logical adjacencies on P2P, P2MP, virtual links. Elect DR/BDR on broadcast and NBMA networks. Continuously send hello packets to maintain bidirectional connectivity, failure to receive = neighbor down (RouterDeadInterval)
Database Sync Protocol: Sync LSDB using DBD packets.

NEIGHBOR STATES
Down: Initial state, no Hello packets exchanged or after RouterDeadInterval Int. Received Hello packet from neighbor but without its own Router ID yet
2-Way: Seen its own Router ID. DR/BDR election is taking place in NBMA
ExStart: (DBD) Bi-dir comm.; highest Router-ID=master. Determine Init Seq Nr
Exchange: Exchange of DBD packets (LSA headers)
Loading: Missing LSAs are requested. (LSRLSU)
Full: Databases fully synchronized. Normal operating state
OSPF ROUTING AND ECMP
– Each router runs Dijkstra per area: link cost = metric from LSAs (1-65535)
– Perfer more specific match (CIDR), else O > O > I > E1 > N1 > E2 > N2
– Routes added to RIB/FIB based on computed next hops
– ECMP: Modified Dijkstra supports Equal-Cost Multi-Path if multiple paths have same cost → routes added with multiple next-hops for load balancing

IS-IS (IGP)
Widely used by ISPs, Fast convergence, ECMP, Extendable
ES: End-host devices are called End Systems (ES)
IS: Routers are called Intermediate Systems (IS)
CONNECTIONLESS NETWORK SERVICES (CLNS)
CLNS: ISO Layer 3 datagram service; supports CLNP, ES-IS, IS-IS.
CLNP: Connectionless Network Protocol, similar to IP, used in ISO stack
ES-IS: host → router discovery protocol
IS-IS: router → router. Link-state routing protocol
Integrated IS-IS: Allows IP routing with IS-IS

NETWORK SERVICE ACCESS POINT (NSAP)			
Variable in length (up to 20 bytes), Identifies node, not IF. Example from IP: 192.168.1.24	49.001.1.0000.0000.0003.00	Area ID (L1)	System ID (R3)
49.0001.1921.6800.1024.00			NSRL

Network Entity Type (NET): Unique router identifier. NSEL=0. Starts with 49
Authority & Format identifier (AFI): NSAP format-authority that assigned it
Initial Domain Identifier (IDI): Variable len, identifies administrative domain
Domain Specific Part Format Identifier (DFI): Specifies format
NSEL (N-Selector): always 0 for routers (1b)
System ID: Unique per router (often based on Io-P/MAC) (6b)
Area ID: Used for routing hierarchy (like OSPF areas) (AFI+DI+DSP)

IS-IS PACKET TYPES (ALL IN L1 OR L2)
All PDUs contain Header with shared common fields + specific routing-related information (Type, Length and Value (TLV)) used to extend protocol
IIH (Hello): Build/maintain adjacencies. Includes system ID, holding time, prio
Functions: discover, build, maintain
Interval: 10s (default); DIS sends every 3.3s on LANs
Multiplier: Missed Hello limit → Holdtime = Interval × Multiplier(def=3)
Multicast: [L1 01-80-c2-00-00-14] [L2 01-80-c2-00-00-15]
LSP (Link State PDU): Contains topology info including prefixes with costs; Unicast on p2p or flooded throughout the area (OSPF: LSA Type 1)
CSNP (Complete SNRP): Sent by DIS, lists all known LSPs (OSPF: DBD)
PSNP (Partial SNRP): Used to request missing or acknowledge received LSPs

ROUTER TYPES
L1 Router: Within one area; all have same LSPDB. No inter-area routing
L2 Router: Backbone router; all have same LSPDB, Routes between areas
L1/L2 Router: Acts as both; separates LSPDBs, redistributes between levels
ADJACENCIES
Backbone must be contiguous: L1 ↔ L1-L2 ↔ L2
L1: Are addresses match unless configured otherwise
L2: Allside L1, unless router is L1 only, areas must not match

POINT-TO-POINT (NO DIS)
Initialized by receipt of IS-IS, exchange p2p IIHs padded to MTU
CSNP: Sent once at adjacency startup
LSP: Advertises topology changes (link-state info)
PSNP: Acknowledges received LSPs or requests missing ones
BROADCAST/MULTI-ACCESS (OIS REQUIRED)
Not triggered by ISHs, instead broadcast IIHs with all neighbors MAC's
Designated Intermediate System (DIS): Create, update pseudonode LSPs. Flood LSPs. Similar to DR in OSPF. No backup DIS.
DIS Election: Highest priority (0-127) (Cisco def=64) = Highest SNPA MAC
Preemption: Enabled → higher prio router automatically takes over DIS Role
Pseudonode: Carried out by DIS. Multiaccess links. Separate for L1/L2
Passive interfaces: Advertise network prefixes without adjacency forming

PATH SELECTION
Default interface metric = 10. **Lower Metric better:** L1 intra-area > L2 intra-area > Leaked L2 → L1 (internal metric) > L1 external (external metric) > L2 external (external metric) > Leaked L2 → L1 (external metric)

LEVEL 1 ROUTING
– Intra-area routing only (L1 Area like OSPF Totally Stubby Area)
– Router install a default L1 route to nearest L1-L2 for inter-area traffic
L1-L2: Do not advertise L2 routes into L1 area (unless route leaking active)
L1-L2: Set **Attached bit** to signal L2 connectivity to backbone
Distribution Bit: Set to 1 on L2 → L1; blocks re-advertisement L1 → L2.
Route-Leaking: injects a more specific route into L1 to improve routing
LEVEL 2 ROUTING
– Routing between areas (inter-area)
– L1-L2 routes inject L2 into L1 topology
– L1 routes are redistributed into L2 with L1 metric preserved in L2 LSP
IS-IS VS OSPF

Feature	IS-IS	OSPF
Layer	L2 (CLNS)	L3 (IP, proto 89)
Encapsulation	IIH, uses TLVs	IP packets
Hello Type	IIH	Hello packet
Area Model	L1/L2	Backbone + Areas
Metric	Cost (default 10)	Cost (bandwidth)
Router ID	System ID (6B)	32-bit Router ID
Adj. Types	L1, L2, L1/L2	DR/BDR, P2P
LSDB	Per level (L1/L2)	Per area
Scaling	Large-scale ISP core	Enterprise/campus
Routing info	TLVs (flexible)	Fixed LSA types
Type	Link-State	Link-State
Shortest Path calculation	Dijkstra	Dijkstra

BGP (EGP)
Path-vector routing protocol. Stable, flexible, scalable. Based on TCP
IGBP
Routers in same AS, AD=200, more trusted (lower security overhead). AS-Path and Next-hop not modified

SCALABILITY
– iBGP does not re-advertise routes between iBGP peers → full mesh required (caused no loop prevention)
– Session count = $n^{(n-1)/2}$ (e.g., 5 routers = 10 sessions, 10 = 45)
ROUTE REFLECTORS (RR)
Solves iBGP full-mesh scaling by allowing selective route reflection. Clients only peer with RR, unaware they're clients. Only the RR needs special config. From non-client, RR advertises to clients only
From client, RR advertises to all clients + non-clients
From eBGP peer: RR advertises to all clients + non-clients

EBGP
Routers in different ASes, AD=20, stricter policy enforcement. Prepends ASN to AS-Path, modifies next-hop IP. Fails if own ASN in AS-Path. TTL=1
AS-Override: Allows reuse of same ASN across different customer sites (e.g., Swisscom); rewrites ASN to avoid loop detection

CONFIG
next-hop-self fixing iBGP: neighbor (neighbor IP) next-hop-self (overriding)
AUTONOMOUS SYSTEM NUMBERS (ASN)
– Unique ID for each AS; required for Internet routing with BGP
– Private: [Legacy 16-bit 64512-65535] [32-bit 4200000000-4294967294]
BGP PEERING / NEIGHBORS
– Two routers with a BGP TCP session (port 179) are called peers/neighbors
– Each BGP router is a **BGP speaker**
– Supports CIDR, route aggregation; decisions based on policies/rules

PATH ATTRIBUTES
Used for route control and policy enforcement
Well-known mandatory: Always present (e.g., AS-Path, Origin, Next Hop)
Well-known discretionary: Optional but recognized by all (e.g., Local Pref)
Optional transitive: Passed between ASes (e.g., Community, Aggregator)
Optional non-transitive: Not passed across ASes (e.g., MED, Weight)
NLRI: Routing table info: prefix, prefix length, and associated path attributes

BGP MESSAGES
OPEN: Establishes session; includes version, ASN, Hold Time, BGP Identifier, optional params. **Hold Time:** Heartbeat in seconds (Cisco def=180s), reset by KEEPALIVE/UPDATE; 0 = session down. **BGP Identifier:** 32-bit Router-ID, manually set or highest loopback/active IP; used for loop prevention
KEEPALIVE: Sent every 1/3 of Hold Time; ensures neighbor liveness
UPDATE: Advertises new routes, withdraws old ones, or both; includes NLRI (prefix + path attributes); can act as KEEPALIVE
NOTIFICATION: Sent on session error (e.g. hold timer expired); terminates session immediately

BGP NETWORK STATEMENTS
– Purpose: Advertise specific prefixes to BGP peers (does not activate interfaces)
– Prefix must exist exactly in the RIB (from static, connected, or learned route)
– Attributes (e.g., origin, next-hop, MED) depend on how the route exists in RIB
– BGP advertises only the best path for a prefix to peers, even if multiple exist

BEST PATH CALCULATION
– BGP maintains all received paths per prefix but advertises only the best one
– Best path is installed in the routing table (only small i path, NH 0.0.0.0)
– Next-hop reachability change
– Interface failure to eBGP peer
– Redistribution change
– New/withdrawn path received
Influence:
– Outbound BGP policy → inbound traffic behavior
– Inbound BGP policy → outbound traffic behavior

BGP BEST PATH SELECTION (IN ORDER):
1) Prefer highest **Weight** (Cisco-specific, local to router)
2) Prefer highest **Local Preference** (global within AS)
3) Prefer routes **advertised by router** (only small i path, NH 0.0.0.0)
4) Prefer **shorter AS path** (only length is compared)
5) Prefer **lowest origin type:** IGP < EGP < Incomplete (I on origin)
6) Prefer **lowest MED** (Multi-Exit Discriminator) (also called metric)
7) Prefer **external (EBGP) over internal (IBGP)**
8) For eBGP: prefer path with **lowest IGP metric** to next-hop
9) For eBGP: **Prefer oldest** (more stable) path
10) Prefer **lowest BGP router ID**
11) Prefer path from **lowest neighbor IP** address

ROUTE FILTERING
– Filters control which routes are received/advertised
– Used for security, traffic shaping, memory optimization
– Tools: prefix-list (IP), **filter-list** (AS-path), **route-map** (flexible match/set)

BGP COMMUNITIES
– 32-bit optional, transitive tag (e.g. **ASN-value, 65000-100**)
– Router install a default L1 route to nearest L1-L2 for inter-area traffic
– Can be added, modified, or removed at each hop
PEERING VS. TRANSIT
Transit: ISP provides full reachability (paid relationship)
Peering: ISPs exchange selected routes; equal relationship, usually unpaid
AS Path Filtering: to avoid getting transit: ip as-path access-list 10 permit ^\$

INTERNET EXCHANGE POINT (IXP)
– Facility where networks exchange traffic via BGP peering
– Reduces transit costs, latency, and offloads upstream links
PUBLIC PEERING
– Members peer via shared switch fabric and a **route server**
– Route server distributes routes but stays out of data path (NEXT_HOP unchanged)
– Minimal policy control; one BGP session to route server
– Simplified setup (one legal contract)
PRIVATE PEERING
– Direct BGP sessions between two parties (1:1)
– May use public or private interconnects
– Full policy control per neighbor
– Requires one session and legal contract per peer

Examples: Equinix, SwissIX (non-profit)
ENTERPRISE CONNECTIVITY OPTIONS
Single-Homed: One ISP; one link (BGP or static); simple but no redundancy
Dual-Homed: One ISP; two links (or routers); redundancy within same provider
Multihomed: Multiple ISPs; improved redundancy and routing control, but avoid being a transit – advertise only customer-owned prefixes
Dual-Multihomed: Multihomed, but two links per ISP
TRAFFIC ENGINEERING (TE)
Outbound TE (Local Pref): Set higher local pref to prefer exit path; affects outbound traffic, highest wins
Inbound TE (AS-Path): Signal entry preference with MED; lowest wins; only works if peer honors it
Inbound TE (AS-Path Prepending): Add own ASN multiple times on backup path; shortest AS-path wins
TE Limitation: AS controls outbound (e.g. local pref); inbound control limited – ISPs may ignore MED

TE with Aggregate: Prefer primary ISP with summarized routes; advertise specific prefixes on backup for failover
Aggregate Impact: Longest-match wins → specific prefixes may steer traffic to alternate ISP; avoid provider-owned aggregates
RPKI – RESOURCE PUBLIC KEY INFRASTRUCTURE
– Framework to validate that a prefix is legitimately originated by a specific ASN
– Prevents route hijacking and accidental mis-originations (route leaks)
– RPKI validates origin only – not the AS-path
KEY CONCEPTS
Trust Anchors (TAs): Root CAs of the 5 RIRs; issue certs for resource holders
ROA – Route Origin Authorization: Digitally signed object that authorizes AS to announce a prefix (contains AS, prefix that AS can originate, max pref length)
RPKI Validators: Software that downloads, verifies, and stores Validated ROA Payloads (VRPs)

RPKI VALIDATION STATES
Valid: Prefix valid, but ASN mismatch or prefix too long
Invalid: Prefix found, but ASN mismatch or prefix too long
Not Found / Unknown: No matching ROA
VALIDATOR DETAILS
– Use TALs (Trust Anchor Locators) to fetch data from RIRs
– Validate cryptographic signatures (via X.509 certs with RFC 3779)
– Outputs VRPs; invalid objects are discarded
– Update frequency: ~24h (recommended), ~30–60min (practice)
– RRDP (RFC 8182) replacing rsync (uses HTTPS)
BGP MONITORING
Event tracking, BGP hijack detection, Route leak detection, RPKI status check, Reachability tracking, AS path change tracking, AS path visualization
NETWORK DESIGN
Pillars: Scalability, Speed, Availability, Security, Manageability → overall Cost
AVAILABILITY CONCEPTS (MOST IMPORTANT REQUIREMENT)
MTBF: Mean Time Between Failures, **MTTR:** Mean Time to Repair
 $A = \frac{MTBF}{MTTR} \times \frac{MTTR}{MTBF} \times MTTR$ MTBF combined: $(\sum_{n=1}^N \frac{1}{MTBF_n})^{-1}$ parallel:
 $\sum_{n=1}^N \frac{1}{MTBF_n}$
– Lower **MTTR** + higher **MTBF** = more availability
REDUNDANCY
– Add reliability, decreases MTBF but increases MTTR and complexity
– Balance: resilience vs. manageability
Backup Paths: Duplicate devices/links on primary path, build extra links for redundancy, consider backup link capacity, consider failover speed
Load Balancing: ECMP, EtherChannel, Port-Channel
HIERARCHICAL DESIGN
Access: Connect end devices, high port count, port security, L2, QoS marking
Distribution: L3, policy control, HSRP/VRRP, loop protection, small fault domain
Core: High-speed backbone, L3 only, no policy, scalable/redundant, no security
Collapsed Core: Combines Core + Distribution (small/medium networks)
FLAT DESIGN
– Any-to-any; small networks, MPLS/LAN setups
– Lacks scalability and control
FABRIC DESIGN
Modern design using **Underlay/Overlay**
Underlay: transport (e.g., IP, MPLS)
Overlay: logical virtual topology (e.g., EVPN)
ENTERPRISE CAMPUS
– 100s/1000s of users, multiple buildings, one physical location
– Multiple interconnected LANs, connected via Ethernet and Wireless
FHRP – FIRST HOP REDUNDANCY PROTOCOLS
– Ensures default gateway is always reachable, PCs can only have one
– Enables fast failover during router failure
VRRP – Virtual Router Redundancy Protocol (Multivendor): Shared virtual IP, real MACs per router, Master router handles forwarding
HSRP – Hot Standby Router Protocol (Cisco): Shared virtual IP + virtual MAC, One active, others in standby → faster
GLBP – Gateway Load Balancing Protocol (Cisco): Load balancing + redundancy, Shared virtual IP + multiple virtual MACs, **ROLES:** AVF: Answers ARP and sends virtual MAC addresses of AVFs
DATA CENTER DESIGNS
ToR – Top of Rack: switches per rack, less cabling, easy expansions/exchanges per rack; scalable glass fiber, ideal for high-speed density (full racks). But more switches, more ports, more L2 Srv-2 Srv traffic, more STP to be managed
EoR – End of Row: 1 switch per row, less switches, higher utilization of ports, switches all at one place, better L2 availability between racks. But more cabling
TRAFFIC DIRECTIONS
North-South: Between external networks (client-server, in/out DC)
East-West: Within DC (e.g., server-server, storage)
THREE-TIER DC
– Access – Aggregation – Core (like Hierarchical)
– Optimized for North-South traffic
– Not ideal for East-West communication
LEAF-SPINE ARCHITECTURE
– Two-tier, Leaf switches (access) connect to Spines (core)
– High performance, low latency
– Scalable, ideal for East-West traffic
MULTICAST
Use-cases 1-to-Many: Streaming, software updates, music-on-hold
Use-cases Many-to-Many: Gaming, VR, stock data, group chat
Benefits: Efficient bandwidth, lower server/CPU load, no redundancy, supports multiport apps
Properties: UDP-based (no delivery guarantee, congestion control, or ordering) Apps must handle drops, duplicates, out-of-order packets
Source: Sends to group IP; doesn't need to join
Receiver: Must explicitly join group to receive traffic
MULTICAST ADDRESS RANGES
224.0.0.0 – 224.0.0.255: Link-Local, TTL = 1 (not forwarded by routers)
224.0.1.0 – 224.0.1.255: Reserved by IANA, routable
232.0.0.0 – 232.255.255.255: Source-Specific Multicast (SSM)
239.0.0.0 – 239.255.255.255: Administratively scoped (private multicast space)
BROADCAST BASICS
L3 routes between subnets; L2 floods within same subnet
L2 (Bridging): MAC flff.flff.flff; switches flood to all ports in VLAN
L3 (Routing):
– 255.255.255.255: local broadcast, not routed
– Directed broadcast (e.g. 10.1.1.1.255): can be routed if enabled
L2 MULTICAST: MAC MAPPING
1) **Get IP:** Example multicast IP address: 239.5.5.5
2) **Convert to Binary:** 239.5.5.5 = 11001111.00001101.00000101.00000101
→ Take only the last 23 bits: 00001101.00000101.00000101
3) **Map to MAC Prefix:** Use fixed MAC multicast prefix: 0100:SE → Map last 23 bits to: 0100:SE.05.05.05
4) **Final MAC Address:** 0100:SE05.0505
IGMP – INTERNET GROUP MANAGEMENT PROTOCOL (HOST TO FIRST-HOP-ROUTER)
Purpose: Manages group membership for IPv4 multicast on each segment
IGMPv1:
– Basic join via query-response mechanism

– No way for a host to leave a group explicitly
– Router sends general membership queries every 60s to 224.0.0.1
– If no report is received, router removes group after timeout
– Receiver has no knowledge of the multicast source
IGMPv2:
– Adds **Leave Group** message (fast pruning of unused traffic)
– General queries to 224.0.0.1 every 125s 'Any hosts interested in any groups?'
– Supports **Group-Specific Queries** e.g. when someone leaves group (anyone still interested in group xy?), reducing broadcast overhead
– Still source-agnostic: receivers don't know who the source is
IGMPv3:
– Adds **source filtering** (Include/Exclude lists)
– Enables **Source-Specific Multicast (SSM)** – receiver requests traffic only from selected source(s), no need for Rendezvous Point (RP) anymore
– Adds support for application-level access control and filtering
– Can also be used in ASM (Any Source Multicast), but mainly with SSM

IGMP SNOOPING
Without snooping: multicast = broadcast on VLAN
With snooping: switch listens to IGMP messages and builds a forwarding table
Default: snooping is enabled; switch needs IGMP Query to operate
REVERSE PATH FORWARDING (RPF)
RPF Check: To avoid loops, verifies that a multicast packet arrives on the interface that a unicast packet destined for the multicast source would be forwarded out of.
RENDEZVOUS POINT (RP)
– Used in Shared Tree (*,G) setups with PIM-SM
– RP acts as the common meeting point for sources and receivers
– RPF check is performed toward the RP (not the source)
– Once the source is known, routers may switch to a Source Tree (S,G)

SHARED TREE VS. SOURCE-BASED TREE
Shared Tree (*,G):
– IGMP host sends a membership report (IGMP Join)
– Router adds (*,G) entry to multicast routing table
– means 'any source' → source is unknown/unspecified
Source-Based Tree (S,G):
– Built when router receives an (S,G) join/report from IGMP host
– s = known multicast source; G = group
– Router adds (S,G) to mroute table once source is known
PTM – PROTOCOL INDEPENDENT MULTICAST (ONLY BETWEEN ROUTERS)
– Relies entirely on the unicast routing table (RIB) for multicast forwarding decisions
– Protocol-independent: works with static routes, OSPF, IS-IS, etc.
PTM-DH – DENSE MODE (NO RP)
Push Model: Floods multicast traffic to all interfaces; then prunes where no receivers exist.

1) **Flooding:** Source sends traffic → forwarded out all multicast-enabled links using unicast RIB
2) **Distribution Tree:** Initially includes entire network (shared tree rooted at source)
3) **Prune Messages:** Routers without interested receivers send prunes upstream to remove themselves from the tree
4) **State Maintenance:** Routers track source, receivers, interfaces to forward/prune per group
PTM-SM – SPARSE MODE (ASM, SSM)
Push Model: Multicast traffic is only sent where requested. Works with IGMP to detect interested receivers and uses unicast routing for forwarding.
1) **Join/Prune:** Routers send explicit join/prune messages to request or stop receiving multicast for group (G) to other routers
2) **Forwarding:** Routers only forward multicast packets for group (G) on interfaces from which explicit joins were received
ASM – ANY SOURCE MULTICAST (HOW IT WORKS WITH RP)
– Works with IGMPv1 or IGMPv2 → receiver does not know the source
– Receiver sends **IGMP Join (*,G)** to its first-hop router
– First-hop router forwards **PIM Join (*,G)** hop-by-hop toward the Rendezvous Point (RP)
– RP acts as a common meeting point for sources and receivers
– Sources send multicast traffic to the RP via a **PIM Register** tunnel
– All routers in the multicast domain must know the RP location
SSM – SOURCE SPECIFIC MULTICAST
– Receiver subscribes using IGMPv3, providing both source (S) and group (G) to the first-hop router
– No Rendezvous Point (RP) required → **PIM-SSM builds only (S,G)** Shortest Path Trees (SPT)
– **No shared tree (*,G)** model used; SSM is source-directed
– IANA reserved 232.0.0.0/8 for SSM in IPv4
– Join messages are forwarded hop-by-hop toward the source to establish forwarding path
– Uses unicast routing table (RPF) to maintain loop-free delivery

PTM SPARSE-DENSE MODE
PIM Sparse Mode: Full mode, multicast traffic is forwarded only on request
PIM Dense Mode: Push model – traffic is flooded everywhere, then pruned
Sparse-Dense Mode: Supports both modes per multicast group; choice depends on RP availability
PTM-DH VS. PIM-SM
PIM is protocol-independent; it relies on the unicast routing table for RPF checks and to forward joins toward the source or Rendezvous Point (RP).
– **PIM Dense Mode (DM)** – **PIM Sparse Mode (SM)**
– "Push" model – "Pull" model
– Floods multicast traffic throughout the network
– Traffic sent only on request
– Requires explicit **Join** messages
– Prunes back where traffic unwanted

Usage Recommendation:
Dense Mode: Best for small or tightly scoped networks where most devices need multicast
Sparse Mode: Preferred for large-scale or distributed environments where multicast receivers are few or spread out
VXLAN
Issues of L2: STP; Max amount of VLANs (4094), Large MAC Address tables
VXLAN (Virtual Extensible LAN): Tunnels Ethernet (Layer 2) over IP using MAC-in-UDP encapsulation (Port 4789). For flexible and scalable network segmentation.

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TODO'S		
OSPF		
– passive interfaces		
– flooding		
– O N1 / O N2 routes		
– route summarization		
– synchronizing the Isdb		
IS-IS		
– ES-IS, IS-IS: details		
– adjacencies		
– route leaking		
– summarization		
BGP		
– comparison to IGP		
– Multihop sessions		
– NLRI		
TODO:		
notes 37+		